CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 3: B-Tree & Hash Index Optimization on High-Volume Tables

Course: Advanced Database Engineering Laboratory [23CS3506]
Category: Experiments / Practicals | Experiment #3 | Academic Year: 2026-27

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1. Overview & Objective:

Benchmark sequential table scans against clustered B-Tree and Hash indexes using execution timer measurements.

2. Practical Tasks & Verification Steps:

1. Populate test table with 1,000,000 synthetic transaction records.
2. Measure baseline sequential scan execution time for equality query WHERE transaction_id = 952341.
3. Create standard B-Tree index and measure query execution latency reduction.
4. Create Hash index for exact equality lookups and compare index storage footprint vs B-Tree.

Submission Guidelines & Verification:

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